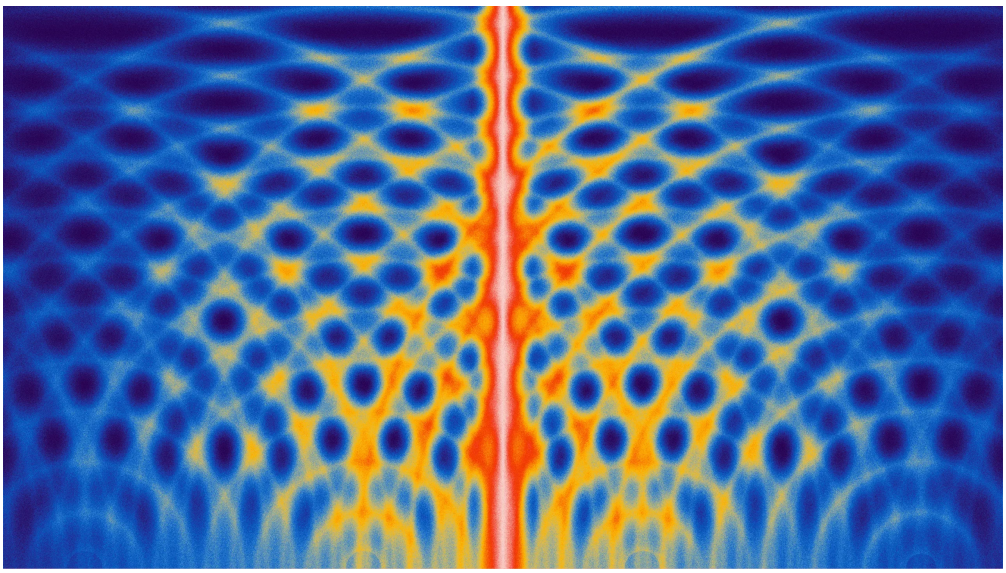
University of California San Diego
Department of Physics
Physics 142/242, Fall 2026
Computational Physics: Stochastic Methods and Path Integrals (4 units)



Note on the course title: the current catalog title and description for PHYS 142/242 ("Computational Physics II: PDE and Matrix Models") predate the way this course and PHYS 141/241 have been taught in recent years. The department is in the process of realigning the catalog descriptions with course content.

Instructor: Javier Duarte, jduarte@ucsd.edu, OH 2-3pm F and by appt.
Teaching assistant: Anthony Aportela, aaportel@ucsd.edu, OH during/after lab TuTh

Course webpage: https://jduarte.physics.ucsd.edu/phys142.
Canvas: https://canvas.ucsd.edu/courses/78771.
Gradescope: https://www.gradescope.com/courses/1404572

Schedule:

Lecture	MWF	3:00p–3:50p	Coalition Building 123
Lab	TuTh	11:00a–12:20p	Pepper Canyon Hall 121

First lecture: Friday, September 25, 2026.
First lab: Tuesday, September 29, 2026. *Labs begin in Week 1; there is no lab on Thursday, September 24.*

Final exam: There is no final exam in this course. The final report and code are due Friday, December 11, 2026, at 8:00pm.

Textbook: There is no required textbook for this course. At the end of the syllabus, we list a bibliography of textbooks and online resources we will draw from.

Course information: This course is an upper-division undergraduate course and introductory graduate course on computational physics, focusing on *stochastic* methods: Monte Carlo integration, Markov chain Monte Carlo, and the statistical analysis of simulation output. We develop these methods in the setting of Feynman's path integral formulation of quantum mechanics, which turns quantum mechanical problems into high-dimensional integrals that are natural targets for sampling. The course will explore both the theoretical foundations and computer implementations. Students will develop their own code to solve the physics applications. Basic knowledge of calculus, quantum mechanics, Linux, and programming in some language is expected.

The course structure will consist of weekly lectures on conceptual topics, e.g., quantum mechanics and Monte Carlo theory, and lab sections on computational tools, e.g., programming in Python and C/C++. Students will learn how to apply physical reasoning to programming, optimize and debug code, create simulations of physical systems, and—critically for stochastic methods—report results with honest uncertainties. Students will also learn how to use modern tools to efficiently solve scientific computing problems: interpreted (Python) vs. compiled (C/C++) languages and how to link the two. There will be 3 individual homework assignments. There will also be a final project in which students will work in groups.

Relationship to PHYS 141/241: **PHYS 141/241 and PHYS 142/242 are independent companion courses. Neither is a prerequisite for the other, and they may be taken in either order.** In academic year 2026–27, PHYS 142/242 is offered in Fall and PHYS 141/241 is offered in Winter, so many of you will take this course first. Nothing in this course assumes you have taken PHYS 141/241.

The two courses divide computational physics by *methods*:

- **PHYS 141/241 (deterministic methods)**: ODE integrators (Euler, Runge–Kutta, symplectic/leapfrog), stability and accuracy analysis, boundary value problems and shooting, PDE solvers (finite difference, Crank–Nicolson, spectral, finite element), deterministic N-body simulation.
- **PHYS 142/242 (stochastic methods, this course)**: pseudorandom number generation and sampling, Monte Carlo integration and variance reduction (importance sampling, VEGAS), Markov chains, the Metropolis–Hastings algorithm, detailed balance, autocorrelation and statistical error estimation, path integrals, and the imaginary-time mapping between quantum mechanics and statistical mechanics.

Week 9 of this course includes a lecture surveying the deterministic methods covered in PHYS 141/241 and how they relate to what we do here.

Student learning outcomes: Upon successful completion of Physics 142/242, students will be able to:

- Design computer programs to numerically solve physics problems, like the harmonic oscillator using the Feynman path integral approach.
- Estimate and report statistical uncertainties on Monte Carlo results.
- Validate a stochastic calculation against an independent deterministic one and explain any residual discrepancy quantitatively.
- Consider multiple approaches and compare their computational performance, accuracy, and fidelity to physical laws.
- Find and choose the best tool or programming language for the task.
- Use AI coding assistants effectively and critically, and recognize its characteristic failure modes.
- Visualize the solutions.
- Collaborate with peers to tackle complex, realistic problems.
- Present findings, and defend the technical choices behind them.

Grading policy: Your final course grade will be determined according to the following:

- 20% Homework (3 assignments).
- 20% Oral checkpoints (3 individual, during lab).
- 20% Quizzes (4 short in-class quizzes, lowest score dropped).
- 30% Final project.
- 10% Participation (attendance).

Academic integrity and the use of AI tools: Please read the College Policies section of the UCSD's Policy on Integrity of Scholarship. These rules will be enforced.

AI coding assistants (ChatGPT, Claude, Copilot, Cursor, and similar tools) are increasingly being used in computational physics. **You are permitted to use them on homework and projects, but you must disclose their use.** The policy for each type of assessment is below.

Tier	Applies to	Rule
0. No AI	Quizzes, oral checkpoints	No AI tools, devices, or notes beyond what is explicitly permitted. At oral checkpoints you may run your own code, without AI assistants.
1. AI as collaborator	Homework, final project	You may use AI tools freely, including to write code, *provided you disclose it* (see below).

AI use appendix (required): Every homework and project submission must end with a short appendix, no more than half a page, containing:

1. Which tools you used, and for what.
2. The prompts you used, or a share link to the conversation.
3. What the tool got wrong, and how you caught it.

"I did not use AI tools on this assignment" is a complete and perfectly acceptable appendix. The appendix is worth points on each assignment.

The third item is important. Language models may be confidently wrong about numerical physics in different ways. Finding those errors is an integral skill that we will spend a lab on (Lab 10).

Enforcement: Submitting AI-generated work with a false or missing AI use appendix, submitting another person's work as your own, or using unauthorized aids during a quiz or oral checkpoint are all violations and will be reported to the UCSD Office of Academic Integrity, and can result in an automatic "F" for the course at the discretion of the instructor.

Collaboration with your peers is encouraged—discuss concepts, approaches, and debugging freely. But by the time you start writing your own solution to turn in, you should not be looking at anyone else's. Plagiarizing a solution but changing variable names is cheating. Soliciting solutions online (Chegg, Quora, course-answer sites) is cheating.

Any questions on what constitutes an academic integrity violation should be addressed to the instructor.

Homework: There will be 3 homework assignments, submitted as code and a report on Gradescope.

- **Parameters:** Some problems assign numerical parameters (e.g., potential shape, temperature, lattice size, or random seed) derived from your student ID.
- **Random seeds:** Every Monte Carlo result should be reported with the random seed that produced it and an uncertainty. We may download your code and re-run it to reproduce your results.
- **Failure modes:** Some problems may ask you to run a method in a regime where it works badly—a time step that is too large, a chain that is too short, etc.—and to diagnose what went wrong from the output.

Oral checkpoints: Three times during the quarter, you will meet individually with the instructor or TA during lab for about 5 minutes to walk us through your own code. You may be asked to run it, change something, or explain a few lines.

If you wrote and understand your code—whatever tools helped you get there—this should be a brief conversation about your own work. They are graded on whether you can explain and modify what you submitted. Checkpoints are tied to Assignments 1, 2, and 3, and are scheduled in Labs 07, 13, and 15.

Quizzes: There will be 4 short quizzes, taken **in class, on paper, closed-device**, each 30 minutes. They cover conceptual material from the preceding lectures. A basic calculator is permitted; phones and laptops are not. Your lowest quiz score is dropped.

Final project: For the final project, students will work in groups of ∼4. The project deliverables are: (1) code provided as a public GitHub repository, (2) a 20-minute presentation by all members of the group, (3) a ∼4-page report in the Phys. Rev. Lett. two-column format (page limit does not apply to references and appendices), and (4) self and peer evaluations for group contributions.

Two notes:

- **Q&A:** Following each group presentation, there will be about 5 minutes of questions in which each group member should be able to answer questions about any part of the project. Divide the work, but make sure everyone understands the whole project.
- **Commit history:** Your repository should show the actual development of the project.

Attendance and participation (lectures and labs): In-person attendance will be recorded. We will use Jibble to track attendance. Join here: https://www.jibble.io/quick-join?magicLinkId=00e54ea39-4973-aa8e-54eca0c4ff8e. You will receive full participation credit if you attend at least 80% of lectures and labs. Lecture will be mostly conceptual, while labs include hands-on portions with interactive problem-solving and pair programming. Quizzes are given in lecture and oral checkpoints are held in lab. Sessions will be recorded.

Exit tickets: At the end of each class, you will be invited to fill out an exit ticket: https://forms.gle/rXb8ZqxLxRrdswQX7.

Discussion board: We will use Discord: https://discord.com/invite/WnDCU6xsGk.

Counseling and Psychological Services (CAPS): The mission of CAPS is to promote the personal, social, and emotional growth of students. Many services are available to UCSD students including individual, couples, and family counseling, groups, workshops, and forums, consultations and outreach, psychiatry, and peer education. To make an appointment, call (858) 534-3755. For more information, visit https://wellness.ucsd.edu/caps/.

Schedule (Subject to change):

Week 0

Thursday 9/24: *No lab* (labs begin in Week 1)

Friday 9/25: Lecture 01: Course overview, the double-slit experiment, and how this course handles AI

Week 1

Monday 9/28: Lecture 02: Quantum mechanics refresher: states, operators, and the propagator

Tuesday 9/29: Lab 01: Python, Jupyter, and DataHub

Wednesday 9/30: Lecture 03: Feynman path integral

Thursday 10/1: Lecture 04: Feynman path integral (continued)

Friday 10/2: Lab 02: NumPy, SciPy, and Matplotlib

Week 2

Monday 10/5: Lecture 05: Free particle; **Quiz 1**

Tuesday 10/6: Lab 03: Random numbers, seeding, and reproducible simulation

Wednesday 10/7: Lecture 06: Harmonic oscillator

Thursday 10/8: Lab 04: Git and GitHub

Friday 10/9: Lecture 07: Schrödinger equation

Week 3

Monday 10/12: Lecture 08: Unitarity and the propagator trace; **Quiz 2**

Tuesday 10/13: Lab 05: Assignment 1 tutorial

Wednesday 10/14: Lecture 09: Unitarity and the propagator trace (continued)

Thursday 10/15: Lab 06: Assignment 1 help

Friday 10/16: Lecture 10: Double well potential

Week 4

Monday 10/19: Lecture 11: The eigenvalue problem as a validation tool (with demo); **Assignment 1 due**

Tuesday 10/20: Lab 07: **Oral checkpoint 1** (Assignment 1)

Wednesday 10/21: Lecture 12: Statistical mechanics recap

Thursday 10/22: Lab 08: Compiled code and Python interop (C/C++, Julia, Numba)

Friday 10/23: Lecture 13: Density matrix and the path integral

Week 5

Monday 10/26: Lecture 14: Markov chain Monte Carlo and the Metropolis algorithm; **Quiz 3**

Tuesday 10/27: Lab 09: Assignment 2 tutorial

Wednesday 10/28: Lecture 15: MCMC (continued): detailed balance, asymmetric proposals, ergodicity, burn-in

Thursday 10/29: Lab 10: Auditing AI-generated numerical code

Friday 10/30: Lecture 16: Autocorrelation and statistical error estimation

Week 6

Weeks 6–7 schedule adjustment: Labs on Wednesday 11/4 and Friday 11/13 meet from 3:00–3:50pm in Coalition Building 123. Lectures on Thursday 11/5 and Tuesday 11/10 meet from 11:00am–12:20pm in Pepper Canyon Hall 121.

Monday 11/2: Lecture 17: MCMC for the 2D Ising model
Tuesday 11/3: Lab 11: MCMC and the 2D Ising model
Wednesday 11/4: Lab 12: Statistical error estimation: autocorrelation, blocking, jackknife, bootstrap; **Assignment 2 due**
Thursday 11/5: Lecture 18: MCMC for the Feynman path integral
Friday 11/6: Lecture 19: MCMC for the Feynman path integral (continued)

Week 7

Monday 11/9: Lecture 20: Assignment 3 overview; **Quiz 4**
Tuesday 11/10: Lecture 21: MCMC for protein folding
Wednesday 11/11: **Veterans Day: No class**
Thursday 11/12: Lab 13: **Oral checkpoint 2** (Assignment 2)
Friday 11/13: Lab 14: Path-integral MCMC tutorial

Week 8

Monday 11/16: Lecture 22: Final project overview; **Assignment 3 due**
Tuesday 11/17: Lab 15: **Oral checkpoint 3** (Assignment 3)
Wednesday 11/18: Lecture 23: Particle physics MC and VEGAS
Thursday 11/19: Lab 16: VEGAS tutorial
Friday 11/20: Lecture 24: Particle physics MC and VEGAS (continued)

Week 9

Monday 11/23: Lecture 25: Companion methods: what PHYS 141/241 covers
Tuesday 11/24: Lab 17: Final project help
Wednesday 11/25: Lecture 26: Final project work session
Thursday 11/26: **Thanksgiving: No class**
Friday 11/27: **Thanksgiving: No class**

Week 10

Monday 11/30: Lecture 27: Final presentations
Tuesday 12/1: Lab 18: Final presentations
Wednesday 12/2: Lecture 28: Final presentations
Thursday 12/3: Lab 19: Final presentations
Friday 12/4: Lecture 29: Final presentations

Finals Week

Friday 12/11: **Final report and code due, 8:00pm**

Bibliography:

[1] R. P. Feynman, "Space-time approach to nonrelativistic quantum mechanics", Rev. Mod. Phys. **20**, 367 (1948).

[2] G. P. Lepage, "A New Algorithm for Adaptive Multidimensional Integration", J. Comput. Phys. **27**, 192 (1978).

[3] K. Banerjee and S. P. Bhatnagar, "Two Well Oscillator", Phys. Rev. D **18**, 4767 (1978).

[4] R. P. Feynman, A. R. Hibbs, and D. F. Styer, *Quantum Mechanics and Path Integrals* (Dover Publications, 2010).

[5] R. Feynman, R. Leighton, and M. Sands, *The Feynman Lectures on Physics, Vol. III: The New Millennium Edition: Quantum Mechanics* (Basic Books, 2011).

[6] P. Eastman, *Introduction to statistical mechanics*, https://web.stanford.edu/~peastman/statmech/phasetransitions.html, 2014.

[7] R. Rodgers and L. Raes, *Monte Carlo simulations of harmonic and anharmonic oscillators in discrete Euclidean time*, DESY Summer Student Programme (2014).

[8] R. H. Landau, M. J. Páez, and C. C. Bordeianu, *Computational Physics: Problem Solving with Python*, 3rd ed. (Wiley, 2015).

[9] F. Krauss and D. Maitre, *Phase transitions*, https://www.ippp.dur.ac.uk/~krauss/Lectures/NumericalMethods/PhaseTransitions/Lecture/pt1.html, 2018.

[10] A. Papaefstathiou, "How-to: write a parton-level Monte Carlo particle physics event generator", Eur. Phys. J. Plus **135**, 497 (2020).

[11] S. Mittal, M. J. Westbroek, P. R. King, and D. D. Vvedensky, "Path integral Monte Carlo method for the quantum anharmonic oscillator", Eur. J. Phys. **41**, 055401 (2020).

[12] G. P. Lepage, "Adaptive multidimensional integration: VEGAS enhanced", J. Comput. Phys. **439**, 110386 (2021).

[13] Y. D. Chong, *Computational physics*, https://phys.libretexts.org/@go/page/34507 (Nanyang Technological University, 2021).